Superstore Sales Analysis - Executive Summary

ApexPlanet Data Analytics Internship - Task 5 Final Report

Executive Summary

Over the four-year period from January 2014 to December 2017 the Superstore recorded **$2,297,200.86** in total sales and **$286,397.02** in profit across 5,009 orders placed by 793 customers, a 12.47% profit margin. The average order was worth $458.61 and carried a 15.62% discount, and profitability turns negative once discounts pass the 20% mark. Technology is the strongest category by profit ($145,454.95), the West is the largest region by sales ($725,457.82) while the Central region is the weakest on margin at 7.92%. Demand is strongly seasonal, peaking every November and December, with November 2017 alone reaching $118,447.83. The results point to three practical levers: disciplined discounting, seasonal planning and segment-specific marketing.

Key Performance Indicators

KPI	Value	KPI	Value
Total Sales	**$2,297,200.86**	Customers	**793**
Total Profit	**$286,397.02**	Products	**1,862**
Profit Margin	**12.47%**	Average Order Value	**$458.61**
Orders	**5,009**	Average Discount	**15.62%**

Supporting Charts

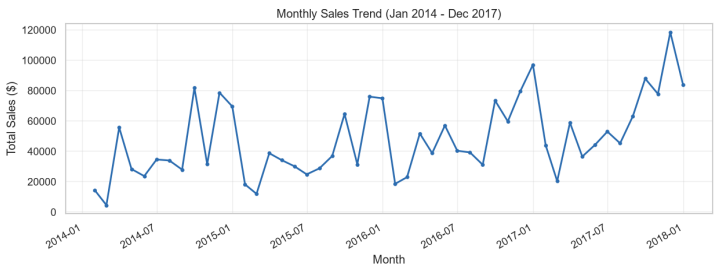


Figure 1 - Monthly sales trend (2014-2017)

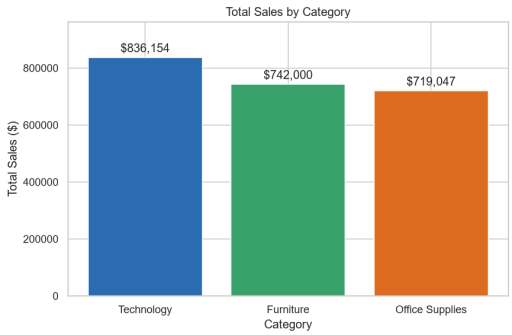


Figure 2 - Total sales by category

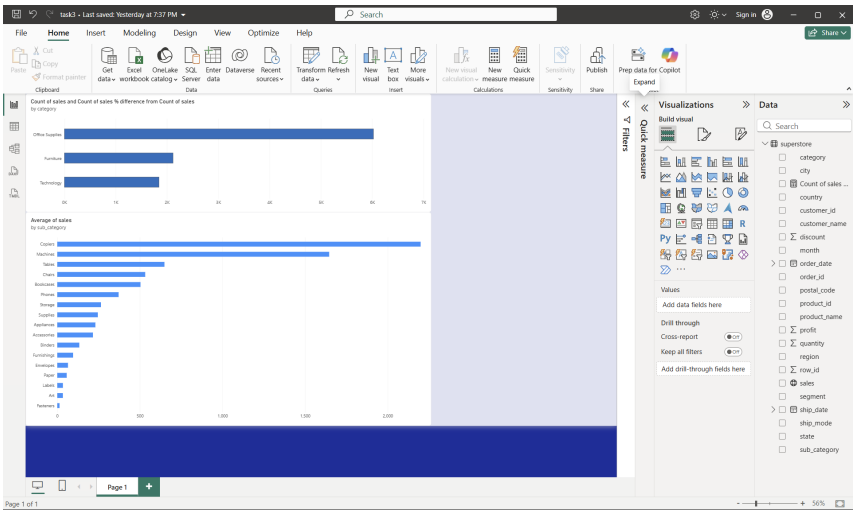


Figure 3 - Power BI dashboard (Task 3)

Insights, Recommendations & Next Steps

Top 5 Insights

1. **Seasonality is strong.** Sales peak every November and December; November 2017 hit $118,447.83 and the seasonal strength score is 0.805 (trend strength 0.279), so demand is far from flat across the year.
2. **Discounting destroys margin.** Profit turns negative above a 20% discount and at 41%+ the margin is -77.40%, losing $99,558.59 in total; discounted lines average -$6.66 profit versus +$66.90 at full price (p ~ 4.4e-55).
3. **Technology is the profit engine.** Technology delivered $145,454.95 of profit, while Furniture's $741,999.80 of sales produced only $18,451.27 - high revenue with thin return.
4. **Customer value is concentrated.** 264 Champions (33.3% of customers) spend $5,024.46 each versus $1,645.11 for Discount seekers; even the top spender, Sean Miller, is unprofitable at -$1,980.74.
5. **Growth and margin live in different regions.** The West is the largest region by sales ($725,457.82) while the Central region is the weakest on margin at just 7.92%, so a single national playbook misallocates effort.

Business Recommendations

1. **Cap routine discounts and reprice the 21%+ bands.** Reserve deep discounts for genuine clearance only, since every point of discount past 20% erodes margin directly (41%+ loses money at a -77.40% margin).
2. **Plan stock, staffing and targets around the Nov/Dec peak.** Budget against the seasonal curve instead of a flat run-rate; this also sharpens forecasting, which is currently limited (3-month moving-average backtest MAPE 50.58%).
3. **Run segment-specific plays.** Loyalty rewards for Champions, cross-sell for the Steady core (34.4% of customers), bundle pricing instead of percentage-off for Discount seekers, and one low-cost reactivation email for the 88 Dormant customers.

Data Source

data/superstore_cleaned.csv (9,994 rows, Jan 2014 - Dec 2017). Generated 2026-10-17 by scripts/automate_pipeline.py.